PhD Candidate Application Interview

Mobile Wireless Sub-THz MIMO Communication for 7G
KTH Royal Institute of Technology

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Overview

1. Academic Background
2. Previous Experience
3. Future Prospect

Academic Background

Academic Background

Education

- **Ingénieur** (2024 – 2026)
  Télécom Paris, France
  GPA: 3.95/4
  Data Science, Large-scale Digital Infrastructure
- **M.Eng** (2022 – 2025)
  Tongji University, China
  GPA: 4.83/5, Rank: 1/48
  Electronic and Information Engineering
- **B.Eng** (2018 – 2022)
  Hangzhou Dianzi University, China
  GPA: 4.72/5, Rank: 1/116
  Information Security

Key Honors

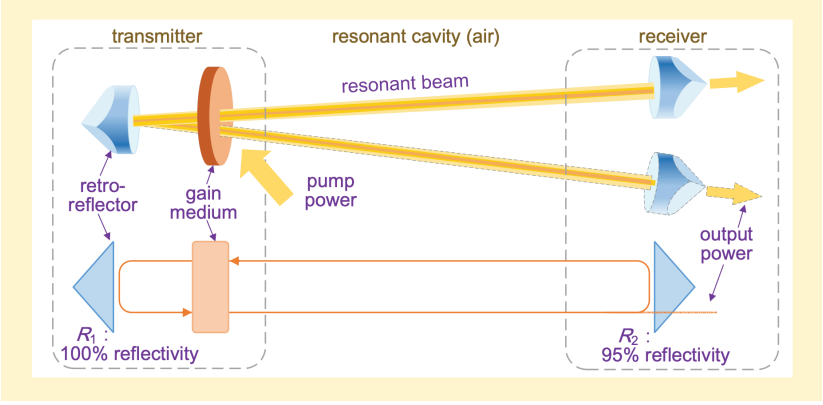
France

- France Excellence Eiffel Scholarship (2024 – 2026)

China

- National Scholarship for Master Students (2023, 2024)
- National Scholarship for Undergraduate Students (2020)
- Top-10 Student of the Year (2021)
- Provincial Scholarship (2019, 2021)

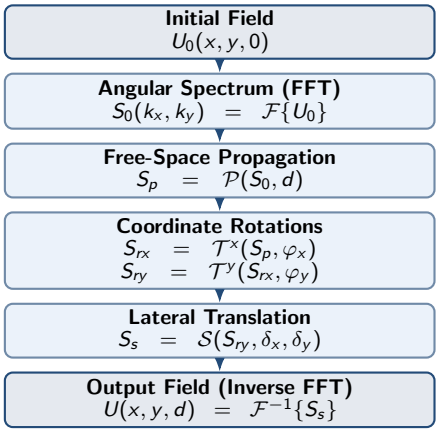
Previous Experience



- Spatially separated laser cavity.
- Adaptive Auto-alignment.

Numerical Modeling & Simulation Framework

Modeling EM Propagation Between Arbitrary Planes



My Contributions

- EM propagation modeling
- Numerical implementation
- Arbitrarily oriented planes

Research Applications

- Beam propagation & focusing
- System configuration analysis
- Numerical validation

From Optical to mmWave Resonant Beam Systems

Research Objective

Extend the optical resonant beam system to mmWave communications, enabling automatic beam alignment without conventional beam training.

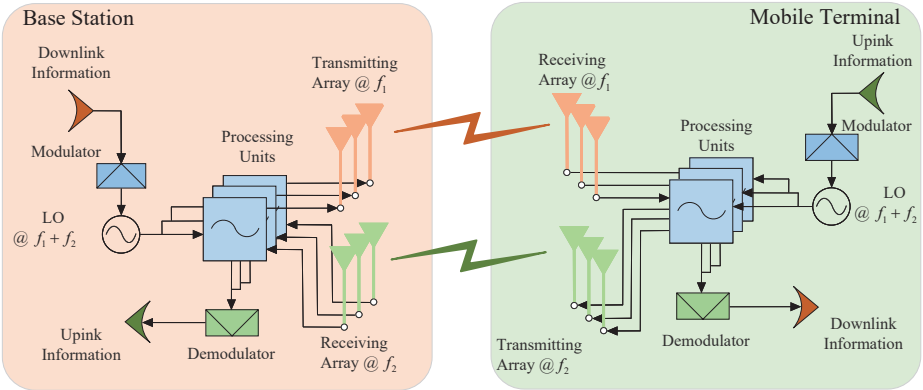
Research Questions

Q1. System Design
How to achieve retro-directivity at mmWave frequencies?

Q2. Mathematical Modeling
How to model EM propagation between antenna arrays?

Q3. Resonance Convergence
How to determine when stable resonance is reached?

mmWave Resonant Beam System



S. Xia et al., "Frequency Division Duplexing Resonant Beam Communication," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 6145–6159, 2026, Fig. 2.

Dual-Frequency Retro-Directivity

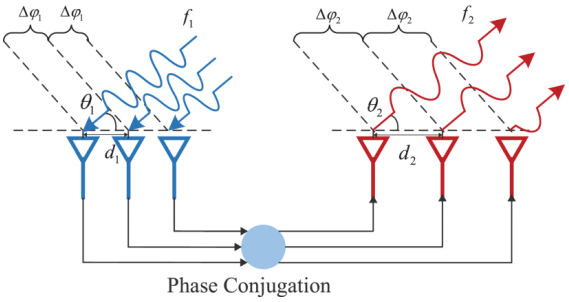
Challenge

- Echo interference between uplink and downlink
- Frequency conversion affects beam direction

Solution

$$\frac{d_1}{\lambda_1} = \frac{d_2}{\lambda_2}$$

Preserve normalized antenna spacing



Maintaining retro-directivity with different frequencies

S. Xia et al., "Millimeter-Wave Resonant Beam SWIPT," *IEEE Internet Things J.*, vol. 11, no. 24, pp. 40464–40477, 2024, Fig. 4.

Modeling Power Propagation

1. Element-to-Element Propagation (Friis' Equation)

$$P_{mn} = P_{T,m} G_{T,mn} G_{R,mn} \left(\frac{\lambda}{4\pi r_{mn}} \right)^2$$

2. Coherent Superposition of EM Fields

$$E_n = \sum_{m=1}^{M} E_{mn}, \qquad P_{R,n} \propto \left| \sum_{m=1}^{M} E_{mn} \right|^2$$

3. Array-Level Received Power & Efficiency

$$P_R = \sum_{n=1}^{N} P_{R,n}, \qquad \eta = \frac{P_R}{P_T}$$

Resonance Convergence: EM Field Perspective

SWIPT Architecture – IoTJ 2024

1. Iterative Propagation

$$\underbrace{E_i^{\mathrm{UD}} \xrightarrow{\text{Harvest, Conjugation, Propagation}} E_i^{\mathrm{BS}} \xrightarrow{\text{Amplification, Conjugation, Propagation}} E_{i+1}^{\mathrm{UD}}}_{\text{Round trip}}$$

2. Recursive EM Field Relationship

$$E_{i+1}^{\mathrm{UD}} = G_T \mathcal{H} \left\{ \left[G_R G_T \mathcal{P} \left\{ \left(G_R E_i^{\mathrm{UD}} \right)^* \right\} \right]^* \right\}$$

$\mathcal{P}$: Power amplification $\quad \mathcal{H}$: Power harvesting $\quad {}^*$: Phase conjugation

3. Self-Reproducing Resonance

$$\boxed{E_{k+1}^{\mathrm{UD}} = E_k^{\mathrm{UD}}}$$

Stable resonance is reached when the EM field reproduces itself.

S. Xia et al., "Millimeter-Wave Resonant Beam SWIPT," *IEEE Internet Things J.*, 2024, Eqs. (21)–(25).

Power Circulation at k-th Iteration

Uplink: $P^{(k)}_{\mathrm{MTTx}} \xrightarrow{\eta^{(k)}_{\mathrm{up}}} P^{(k)}_{\mathrm{BSRx}} \xrightarrow{g_{\mathrm{BS}}(\cdot)} P^{(k)}_{\mathrm{BSTx}}$

Downlink: $P^{(k)}_{\mathrm{BSTx}} \xrightarrow{\eta^{(k)}_{\mathrm{down}}} P^{(k)}_{\mathrm{MTRx}} \xrightarrow{g_{\mathrm{MT}}(\cdot)} P^{(k+1)}_{\mathrm{MTTx}}$

Power Gain and Loss

Total power loss:

$$\Delta^{(k)} = P^{(k)}_{\mathrm{MTTx}} - P^{(k)}_{\mathrm{MTRx}}$$

Total power gain ($k > 1$):

$$G^{(k)} = P^{(k)}_{\mathrm{MTTx}} - P^{(k-1)}_{\mathrm{MTRx}} + P^{(k)}_{\mathrm{BSTx}} - P^{(k)}_{\mathrm{BSRx}}$$

Convergence Criterion

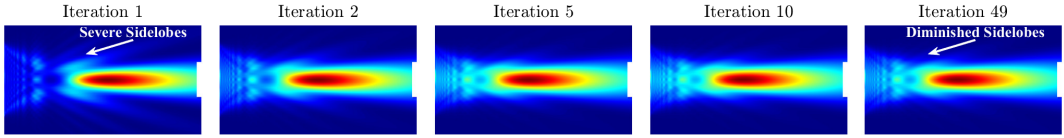
$$G^{(1)} > \Delta^{(1)}$$

$$\lim_{k\to\infty} G^{(k)} = \lim_{k\to\infty} \Delta^{(k)}$$

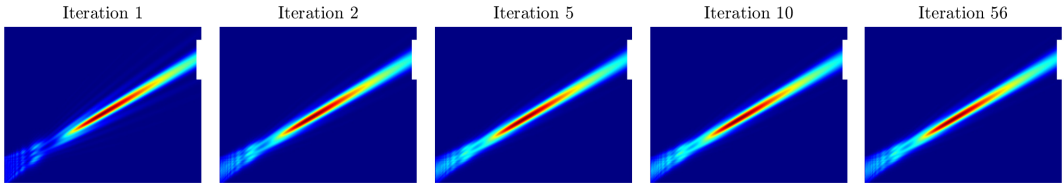
S. Xia et al., "Frequency Division Duplexing Resonant Beam Communication," *IEEE TWC*, 2026, Eqs. (27)–(29).

Simulation Results: Resonance Establishment

Coaxial Configuration



Misaligned Configuration

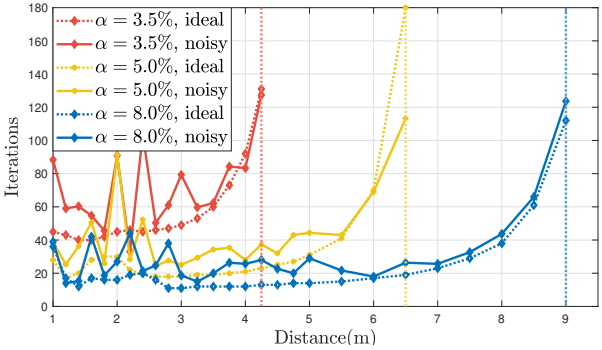


Progressive beam focusing through iterative retro-directive transmission

S. Xia et al., "Frequency Division Duplexing Resonant Beam Communication," *IEEE TWC*, 2026, Fig. 8.

Numerical Evaluation: Resonance Convergence

Convergence Under Different Conditions



Key Observations

Transmission Distance
More iterations at longer distances.

Power Splitting Ratio
Higher ratios extend the operating range.

Phase Noise
Fluctuations and delayed convergence.

Resonance Initialization Time

$$t = \frac{2Ln_{\text{itr}}}{c} \quad \Longrightarrow \quad \sim \mathbf{7.32}\ \mu\mathbf{s}$$

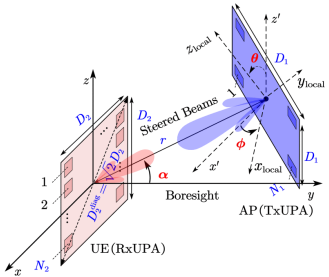
S. Xia et al., *IEEE Trans. Wireless Commun.*, 2026, Fig. 9(a), Eq. (43).

Reflections & Limitations

Limitations of Previous Work

- Primarily theoretical analysis and numerical simulations
- No complete RF hardware prototype
- Far-field assumptions deserve further examination

Related Research at KTH



Array geometry and misalignment affect near-field boundaries.

A Question Worth Revisiting

How would near-field propagation affect resonance convergence?

P. Zhang, V. Petrov, and E. Björnson, "Near-Field Boundary Distance in mmWave and THz Communications With Misaligned Antenna Arrays," *IEEE TWC*, 2026.

Future Prospect

Future Research Interests

Mobility-Aware Beam Management

- Efficient beam alignment and tracking
- Robust communication under user mobility
- Reduced beam-training overhead

Potential Research Approaches

- Wireless propagation modeling
- Optimization and machine learning
- Adaptive beamforming strategies

From Retro-Directive Self-Alignment to Mobility-Aware Beam Management

Why KTH & Future Development

Relevant Research Experience

- Experience in wireless propagation modeling and beam alignment
- Numerical simulation and computational methods
- Interdisciplinary research experience

Areas for Further Development

- Advanced MIMO theory and signal processing
- Practical RF system implementation and experimental validation

Long-Term Goal

Continue working on advanced wireless communication research

Thank You for Listening

Selected Publications

First-Author Publications

1. S. Xia, Q. Liu, Q. Jiang, W. Fang and M. Liu, "**Frequency Division Duplexing Resonant Beam Communication**," in *IEEE Transactions on Wireless Communications*, vol. 25, pp. 6145-6159, 2026.

   Contribution: System design and theoretical modeling, simulation development, and experimental validation.

2. S. Xia et al., "**Millimeter-Wave Resonant Beam SWIPT**," in I*EEE Internet of Things Journal*, vol. 11, no. 24, pp. 40464-40477, 15 Dec., 2024.

   Contribution: System design and theoretical modeling, simulation development, and experimental validation.

3. S. Xia et al., "**Auto-Protection for Resonant Beam SWIPT in Portable Applications**," in *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 4127-4138, 1 Feb., 2024.

   Contribution: System design and theoretical modeling, simulation development, and experimental validation.

Selected Publications

Major Contributor Publications

1. X. Wang, S. Xia, W. Fang, M. Liu, M. Xiong and Q. Liu, "**Mobile Self-Protection Resonant Beam SWIPT With Adaptive Phase Control**," in *IEEE Internet of Things Journal*, vol. 12, no. 20, pp. 43103-43118, 15 Oct., 2025.
   Contribution: Conceptualization, and core methodology proposal.
2. Y. Guo, S. Xia, M. Xiong, Q. Liu, W. Fang, Q. Jiang, G. Yan and J. Mu, "**Integrated Sensing and Communication System Based on Radio Frequency Resonance Beam**," arXiv preprint arXiv:2501.18878.
   Contribution: Conceptualization, and core methodology proposal.
3. M. Liu, S. Xia, M. Xiong, M. Xu, Q. Liu and H. Deng, "**NLOS Transmission Analysis for Mobile SLIPT Using Resonant Beam**," in *IEEE Transactions on Wireless Communications*, vol. 23, no. 1, pp. 334-346, Jan. 2024.
   Contribution: Theoretical derivations and mathematical analysis, simulation development, and experimental validation.
4. M. Liu, S. Xia, M. Xiong, M. Xu, W. Fang and Q. Liu, "**Integrated Communication and Positioning With Resonant Beam**," in *IEEE Transactions on Wireless Communications*, vol. 21, no. 11, pp. 9186-9199, Nov. 2022.
   Contribution: Theoretical derivations and mathematical analysis, simulation development, and experimental

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Contributing Author Publications

1. Y. Guo et al., "**Resonant Beam Multi-target DOA Estimation**," in *IEEE Internet of Things Journal*, vol. 13, no. 10, pp. 20868-20882, 15 May, 2026.

   Contribution: Mathematical modeling, and simulation development.

2. M. Xu et al., "**Individual-Source Resonant-Beam-Enabled 3D Positioning for IoT Scenarios**," in *IEEE Internet of Things Journal*, vol. 11, no. 23, pp. 38673-38685, 1 Dec., 2024.

   Contribution: Simulation development, and experimental validation.

3. S. Han et al., "**Enhanced Field of View for Resonant Beam Systems in IoT Applications**," in *IEEE Internet of Things Journal*, vol. 11, no. 11, pp. 20379-20391, 1 June, 2024.

   Contribution: Simulation development, and experimental validation.